

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

ANSWERS

CLASS

BIOLOGY

9744/03

Paper 3 Long Structured and Free response Questions

September 2017

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper

Section B

Answer any **one** question in the spaces provided on the separate answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	17
2	20
3	13
Section B	
Total	

This document consists of 18 printed pages and no blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 About one third of the injuries to racehorses involve tendon damage. In 2006, bone marrow stem cells were taken from injured racehorses and cultured so that they divided many times by mitosis. Each horse's cells were then injected into its damaged tendons. 80% of the treated horses returned to racing, compared with 30% of those treated conventionally.

- (a) Adult stem cells such as these are described as multipotent.
Explain what is meant by the term multipotent.

multipotent cells can differentiate into a limited number of cell types (but not into whole organism) ;

[1]

- (b) In these treatment procedures, the bone marrow stem cells are stimulated to increase their rates of cell division so as to produce large enough numbers of cells to be injected into the horses.
Using your knowledge of cell signalling, describe how the rate of mitosis can be controlled.

Ref to ligand-receptor interaction:

external growth factors (first messengers) / mitogens / cytokines / signal molecules/
AW ;
receptors on cell (surface) membrane ;

Ref to signal transduction:

activate kinases, kinases phosphorylate enzymes (and alter their activity) ;
ref to idea of a phosphorylation cascade leading to signal amplification;

Ref to response:

activate, transcription factors / promoters ;
increase production of cyclins/ proteins controlling cell cycle/ expression of POGs ;
cyclins / kinases control stages of, cell cycle / mitosis ;

A ref to checkpoints during cell cycle

[4]

- (c) Suggest how it is possible that bone marrow stem cells could differentiate into the range of cell types needed for repairing injuries.

Compulsory point:

stem cells are genetically identical and contains the complete genome/ all the genes;
(example of cell types required to rebuild tendon), blood vessel cells, connective tissue cells, fibroblasts;

different (sets) genes are turned on in different cell types/ different genes will be expressed in different cell types/ under different conditions;

gene expression alters the possibility of expression of further genes;
reference to influence from microenvironment/ influence of signals from adjoining cells as well as from distant cells;

[2]

Like most mammals, T cell (T lymphocytes) in horses differentiate inside the thymus gland. During T cell differentiation, specific cell surface proteins known as CD proteins are produced and inserted into the cell surface membrane.

Fig. 1.1 shows the stages involved in the synthesis of a CD protein in a T cell.

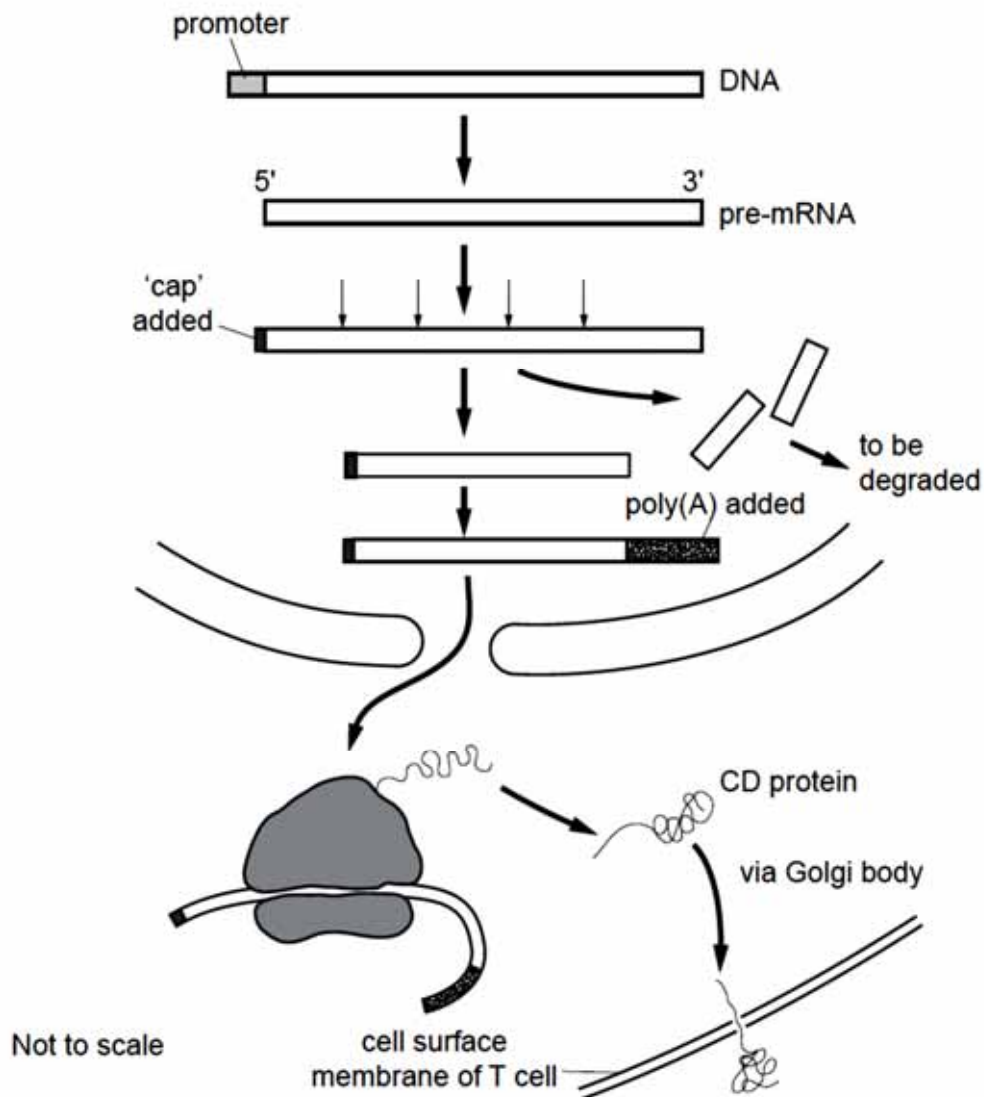


Fig. 1.1

(d) Explain the function of the promoter region of the DNA.

RNA polymerase binds to promoter region ;
transcription factors also required ;
starts transcription ;
upstream of gene to be transcribed ;
ref to at 5' end ;

[2]

- (e) Using **Fig. 1.1** as a guide, describe the events that occur in the nucleus of the T cell to produce a pre-mRNA encoding a CD protein.

transcription/ assembly of nucleotides/ nucleoside triphosphates ;
 RNA polymerase ;
 base pairing / e.g. (A – T / U – A / C – G / G – C) ;
 phosphodiester bonds ;

[3]

- (f) Explain why the cuts made in pre-mRNA are necessary for the T cell to produce a functional CD protein.

introns are non coding
 have to be excised, remaining exons are ligated to produce a continuous coding sequence/ only exons are to be translated;
 ref to correct amino acid sequence that can fold to give the specific 3D structure;

[3]

- (g) Explain the functions for the 'cap' and the poly-A region attached to the mRNA.

prevent breakdown of, mRNA / exons ;
 directs mRNA to ribosome / AW ;

cap identifies 'start' of mRNA ;
 assembly point for small and large subunits of ribosomes ;
 AVP ; e.g. distinguished from viral RNA / identifies host RNA

[2]

[Total: 17]

- 2 Lactate dehydrogenase Lactate dehydrogenase (LDH) is an enzyme found in many organisms. Within the same organism, it can be found in different forms, called isoenzymes. The isoenzymes are structurally different but all catalyse the same reaction.

(a) **Fig. 2.1** shows a reaction catalysed by lactate dehydrogenase that occurs during anaerobic respiration in muscle tissue.

(i) Complete **Fig. 2.1** by identifying the compounds **A**, **B** and **C**.

A: pyruvate;

B: reduced NAD /
NADH + H⁺

C: oxidised NAD /
NAD⁺
B and C one mark

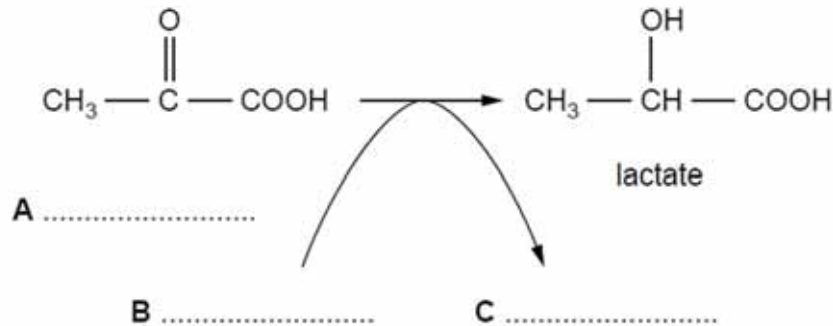


Fig. 2.1

[2]

(ii) State where in the cell this reaction takes place.

Cytoplasm / cytosol;

[1]

(iii) Explain the importance of this reaction in mammalian muscle tissue.

regenerates NAD and allows glycolysis to continue (during oxygen deficit) ;
allows ATP production (to continue) for (muscle) contraction ;

accept details of ATP involvement in contraction:

e.g. temporary storage of hydrogen/ hydrogen transferred prevents accumulation of reduced NAD/AW

e.g. lactate transported areas with (more) oxygen (for oxidation)

e.g. lactate prevents damage to muscles by overexertion/AW

[3]

Lactate dehydrogenase isoenzymes are globular proteins, each consisting of four polypeptides.

(b) Explain how the structure of an enzyme, such as lactate dehydrogenase is suited to its role.

Tertiary/ quaternary structure held in place by, bonds /interactions between R groups ;

Forming specific 3D structure with active site complementary to substrate;

Ref to active site being complementary in terms of shape, size, charged, orientation;

E-S complex formation lowers activation energy for catalysis;

(Globular) protein with hydrophilic amino acids on the exterior allowing protein to be soluble;

(soluble), reactions in aqueous medium/ ref to transport of enzyme;

[4]

Lactate dehydrogenase isoenzymes are made up of two types of polypeptide: polypeptide M, which is coded for by the *LDH-A* gene and polypeptide H, which is coded for by the *LDH-B* gene.

Table 2.1 shows the composition of different human lactate dehydrogenase isoenzymes and examples of tissues and organs where each can be found.

Table 2.1

isoenzyme	polypeptide composition of enzyme	example of isoenzyme location
LDH-1	HHHH	heart red blood cells
LDH-2	HHHM	heart red blood cells
LDH-3	HHMM	brain lungs
LDH-4	HMMM	kidneys placenta
LDH-5	MMMM	liver skeletal muscles

(c) With reference to **Table 2.1**, suggest how red blood cells of the same individual can produce different isoenzymes.

genome of red blood cell contains both genes ;
control of assembly of transcribed polypeptides to produce different proteins/ description
of the different chains that make up the different protein;
different R group interactions between the different polypeptide chains, resulting in
different quaternary structure / specific 3-dimensional structure

[2]

Besides lactate dehydrogenase, another extensively studied protein which is also involved in respiration is cytochrome c. Cytochrome c plays an important role in oxidative phosphorylation as an electron carrier of the electron transport chain.

(d) **Fig. 2.2** shows the amino acid sequence of a section of the cytochrome c polypeptide chain retrieved from a human and the other species. The dashes shown in the figure indicates that the amino acid present at the position is identical to that of the human species.

Molecular homology of cytochrome c																							
		1				6				10				14			17	18		20			
Human		Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His	Thr	Val	Glu	Lys
Pig		-	-	-	-	-	-	-	-	-	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	-
Chicken		-	-	Ile	-	-	-	-	-	-	-	Val	Gln	-	-	-	-	-	-	-	-	-	-
Dogfish		-	-	-	-	-	-	-	-	Val	-	Val	Gln	-	-	Ala	-	-	-	-	-	-	Asn
Drosophila	<<<	-	-	-	-	-	-	-	-	Leu	-	Val	Gln	Arg	-	Ala	-	-	-	-	-	-	Ala
Wheat	<<<	-	Asn	Pro	Asp	Ala	-	Ala	-	-	-	Lys	Thr	-	-	Ala	-	-	-	-	-	Asp	Ala
Yeast	<<<	-	Ser	Ala	Lys	-	-	Ala	Thr	Leu	-	Lys	Thr	Arg	-	Glu	Leu	-	-	-	-	-	-

Fig. 2.2

- (i) Suggest what the data in **Fig. 2.2** indicate about the evolutionary relationships between humans and the other species.

all the species share a (recent) common ancestor ;
 the smaller the number of differences the more closely related/more recently divergence occurred ;
 example + data quoted;
 e.g. pigs and chickens are phylogenetically closest to humans
 e.g. yeast, is the most distantly related to humans (compared to any of the other sp.)
 e.g. number of differences / description of bases at certain positions – see table below

Table of number of differences

	Human	Pig	Chicken	Dogfish	Drosophila	Wheat	Yeast
Human	0	3	3	5	6	10	11
Pig	3	0	2	2	3	9	11
Chicken	3	2	0	4	5	10	11
Dogfish	5	2	4	0	3	10	12
Drosophila	6	3	5	3	0	10	10
Wheat	10	9	10	10	10	0	11
Yeast	11	11	11	12	10	11	0

[3]

- (ii) Explain why any such conclusions in (i) need to be treated with caution.

Ref to cyt being a large protein and only a small fraction is compared:
 the compared sequences are only a small fraction of the whole genome + (figures) e.g. 22 amino acids shown in the table;
 cytochrome is likely to be a larger, polypeptide/protein;

Ref to variation within each species:
 these sequences come from one individual from each species, there will be variation within each species ; A small sample size/ use of amino acid sequences does not allow for detection of silent mutations in the genome due to degeneracy of genetic code;

Ref to changes not fully observed in amino acid sequence:
 amino acid sequence only reflects the coding region of the DNA sequence thus changes in the non-coding regions will not be detected;

[2]

- (e) Suggest how the differences in the amino acid sequences shown in **Fig. 2.2** may have come about.

(point mutation), change in a single base/ substitution ; R addition/ deletion/ frame shift mis-pairing during DNA replication ;
mutagen/named mutagen ; A UV/X-rays / ionising radiation/AW;

[2]

- (f) Suggest why the cytochrome c protein was chosen to compare amino acid sequences across the different species.

Cytochrome c protein is **ubiquitous** / will be present in all organisms because all organisms need to respire which serves as a good basis of comparison between organisms;

Essential / important gene which changes very slowly, useful for estimating time of divergence that occurred long time ago;

accumulates mutations at a constant rate and therefore can be used to calibrate a molecular clock for the estimation of time of divergence between species;

[1]

[Total: 20]

- 3 Dengue fever (DF) is a disease caused by infection with a virus transmitted by the *Aedes aegypti* mosquito. It is an acute viral infection characterized by fever, rash, headache and muscle and joint pain. Occasionally, dengue virus infections progress to dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS). There are 4 known serotypes of the disease-causing dengue virus. Individuals who become infected with one serotype obtain lifelong immunity against that serotype but not the other three.

Female *Aedes* mosquitoes are responsible for human-to-human transmission of the dengue virus. During blood feed, they acquire the proteins necessary for them to develop eggs. As a result of their short life cycle, they are able to multiply quickly, allowing dengue to spread.

- (a) (i) State one reason why blood is a good source of protein.

Blood comprises red blood cells which contain haemoglobin /

Blood plasma contains soluble proteins such as antibodies / protein hormones / clotting factors / enzymes /

Blood has white blood cells that contain / secrete proteins such as major histocompatibility complexes / cytokines / perforins / granzymes ;

AVP

[1]

(ii) Outline the life cycle of the *Aedes aegypti*.

Egg: Female *Aedes aegypti* mosquito lays eggs singly in moist places, such as inner walls of artificial containers, above the waterline;

Larva: When the water level rises (for e.g. rain), and the eggs get submerged in water, the eggs hatch and the larvae emerge after 1 – 2 days ;

Larvae feed on micro-organisms and particulate organic matter and breathe through a siphon at the water surface

The larvae go through larval stages (four instars and moult four times), after which they become pupa that do not feed and will develop into the mosquitoes' adult body form in two days ;

Adult: The newly form adult will emerge after breaking the pupal skin, resting on the water surface to dry and harden their exoskeleton before flying ;

[4]

Aedes aegypti originated in Africa but migrated to other continents via the slave trade in the 1500s and 1600s. The estimated range of the mosquito is primarily the tropics and sub-tropics but they have also been detected in countries with temperate climate such as parts of Europe and North America. As a result of the spread of *Aedes* vector across the globe, worldwide incidences of dengue have also been on the rise, posing huge public health concerns. **Fig. 3.1** shows the annual average number of both DF and DHF reported to the World Health Organisation (WHO) and the average annual number of countries reporting these cases from 1955 to 2007. The upward trend continues today.

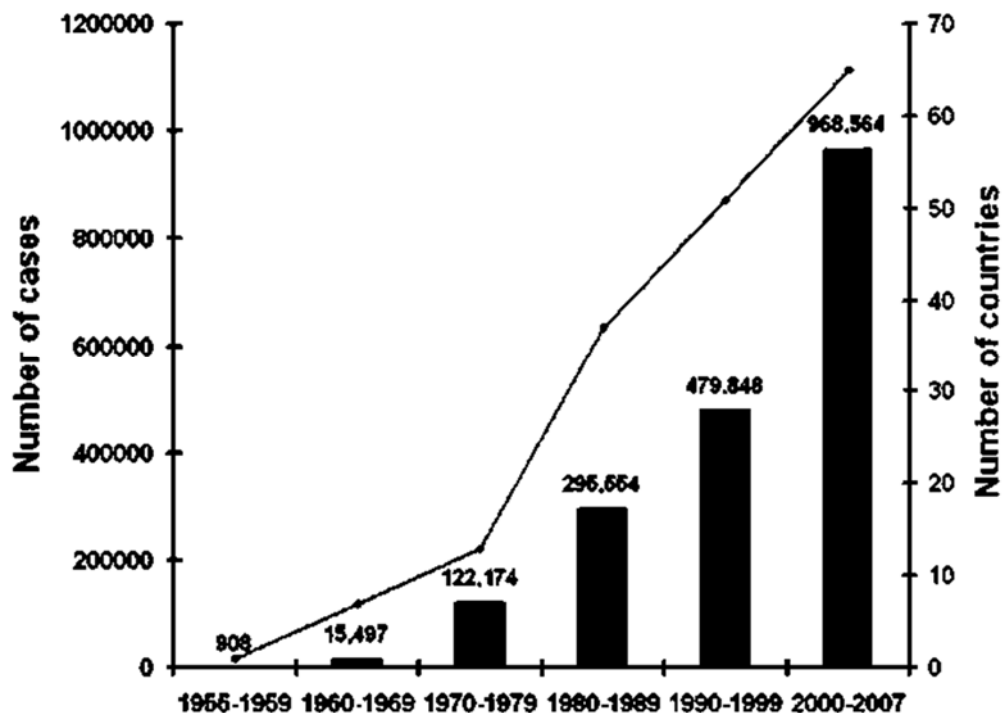


Fig. 3.1

While *increased* human population and global movement of people and cargo via air travel have undoubtedly assisted the spread of dengue, some scientists have attributed it to the effect of global warming.

Fig. 3.2 shows the global temperature departure (°F) from the long-term average from 1880 to 2010. An additional trend line was plotted for the departure values between 1950 and 2010.

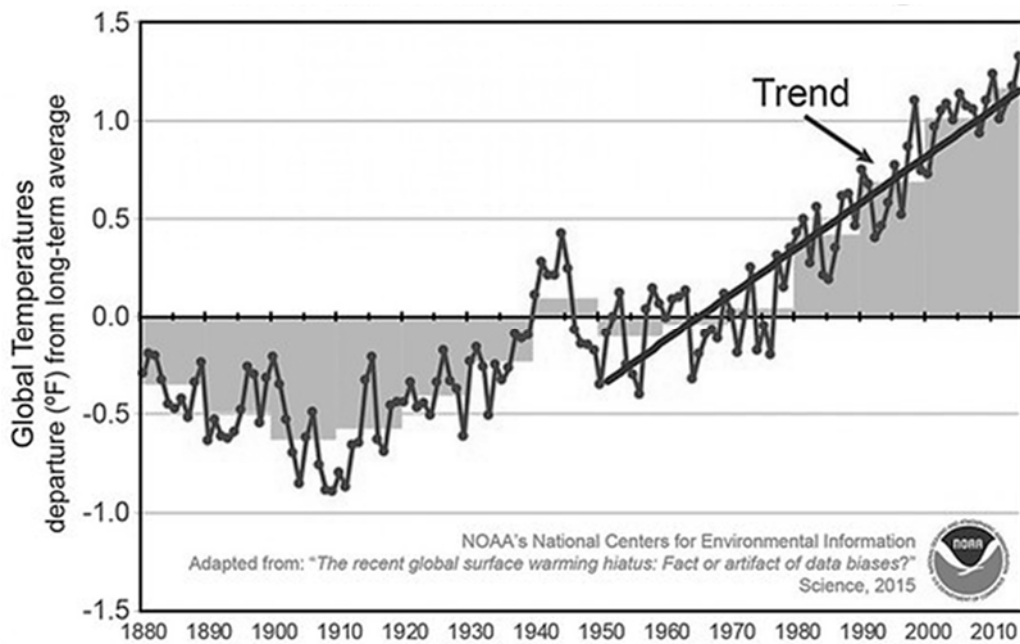


Fig. 3.2

- (b) Using evidence from **Fig 3.1** and **Fig. 3.2** and your own knowledge of global warming,
- (i) explain how the increase in global annual average number of DF / DHF could be caused by global warming.

Hypothesis supported:

As the global temperature departs more and more (positively), from 0°F to 1.2°F away from the long term average (denoted by 0°F), the higher the annual average number of DF / DHF (increase from about 900 in 1955-1959 to about 970,000 in 2000-2007) ;

Explain: Global warming results in increase in global temperatures resulting in faster transmission of dengue virus from any one of the following :

- cold-blooded animals, hence body temperatures could increase with global warming → increase metabolic rate / shorten life cycle → faster growth and development /
- increase activity of female mosquitoes (increase biting rates) /
- increase egg-laying rates and more eggs laid per female /
- reduce the extrinsic incubation period of the virus within mosquito, so mosquito has a greater chance of infecting a human before it dies ;

[3]

- (iii) suggest how global warming can lead to the spread of DF / DHF beyond the tropics into the temperate regions.

Global warming can lead to warmer winter temperatures in the temperate regions: which may lead to the pole-ward range expansion of the mosquitoes, arriving in the now-warmer temperate regions earlier / which may decrease mosquito mortality, causing them to survive the now-warmer winter season in the temperate regions, allowing them to continue multiplying throughout winter ;

(and leading to more rapid transmission of the dengue virus)

[1]

- (c) *Wolbachia* is a natural bacterium present in up to 60% of insect species, but it is not usually found in the *Aedes aegypti* mosquito. Research has shown that when introduced into the *Aedes aegypti* mosquito, *Wolbachia* can stop the dengue virus from replicating inside the mosquito and hence prevent transmission to humans.

One possible way in which *Wolbachia* is used to suppress the *Aedes aegypti* mosquito population is through the release of male *Wolbachia-Aedes aegypti* mosquitoes. When these mosquitoes mate with the female wild-type *Aedes aegypti* that do not have *Wolbachia*, those females will lay eggs that do not hatch.

Country X released these male *Wolbachia-Aedes aegypti* mosquitoes in several housing estates as a form of vector control. Following three months of releases at the study site, the eggs were collected using ovitraps placed at several sites within the estate. The percentage of unhatched eggs were recorded for both the study and control sites.

- (i) State a statistical test that could have been used to determine whether the difference in the mean percentage of unhatched eggs between the study and control sites is significant.

t-test ;

[1]

- (ii) Suggest how a suitable control site was selected for the study.

same population density/ same housing type/ AVP;

[1]

- (iii) A summary of the results is shown in **Table 3.1**.

Table 3.1

mean percentage unhatched eggs		probability
study site	control site	
62.2	15.7	p < 0.00001

Comment on what these results show and suggest an explanation for any pattern.

at 0.05 level of significance, the difference between the mean percentage of unhatched eggs in the study site and the control site is significant ;

The released male *Wolbachia-Aedes aegypti* mosquitoes have successfully mated with the wild-type female *Aedes* ; / control of mosquito using *Wolbachia* is effective;

[2]

[Total: 13]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4(a)** Outline the role of ATP in living organisms using named examples. [13]
(b) Discuss how the survival of species depends on DNA molecules being stable yet not *absolutely* stable. [12]

[Total: 25]

- 5(a)** Outline how variation can arise in any population of living organisms. [13]
(b) Discuss the suggestion that all living organisms on Earth depend on phosphate. [12]

[Total: 25]

QUESTION 4

(a) Outline the role of ATP in living organisms using named examples. [13]

Introduction:

1. Ref to ATP as the universal energy currency / carrier in cells of living organisms;
2. (Explain why) It is small and water soluble, hence can be transported within the cell easily / a lot of chemical energy is stored in the bonds of ATP / hydrolysis of the third phosphate group releases large amount of energy ;

In cellular/aerobic respiration:

3. Activation of glucose to glucose-6-phosphate via phosphorylation using ATP during the energy investment phase of glycolysis. This process is catalysed by hexokinase ;
4. ATP is required for the phosphorylation of fructose-6-phosphate to fructose-1,6-bisphosphate catalysed by phosphofructokinase ;
5. ATP also serves as an allosteric inhibitor to the phosphofructokinase enzyme ;

Uses in cellular activities:

6. Movement of secretory vesicles from the Golgi apparatus to the cell surface membrane along the cytoskeleton requires energy from the hydrolysis of ATP ;
7. Active transport of molecules against concentration gradient across the cell surface membrane via the action of a specific carrier proteins called "pump" which use ATP to change its conformation. E.g. is sodium-potassium pump ;
8. Bulk transport such as endocytosis and exocytosis involves the transport of large molecules like proteins and polysaccharides which require ATP. E.g. secretion of antibodies by plasma cells ;
9. ATP required for amino acid activation prior to translation for the covalent attachment of the amino acid to the 3' acceptor stem of the corresponding tRNA, catalysed by amino acyl tRNA synthetase;

In photosynthesis:

10. Energy from hydrolysis of ATP during Calvin cycle of the light-independent stage of photosynthesis in stroma of chloroplasts in plant cell is required to convert glycerate-3-phosphate (GP) to glyceraldehyde-3-phosphate (GAP) ;
11. Energy from hydrolysis of ATP is also needed to regenerate ribulose biphosphate in Calvin cycle ;

Role of ATP as a nucleotide:

12. ATP is a ribonucleotide and is incorporated into mRNA acid during transcription ;

Role of ATP in cell signalling:

13. ATP is required as a substrate for the adenylyl cyclase enzyme, which upon activation by the activated GTP-bound G-protein, catalyses the conversion of ATP to cAMP, which serves as a second messenger in the signal transduction pathway ;
14. ATP is also the substrate for the kinases in the phosphorylation cascade, which catalyses the addition of phosphate groups from the ATP molecules to the subsequent kinases in the cascade ;

(b) Discuss how the survival of species depends on DNA molecules being stable yet not *absolutely* stable. [12]

DNA as a stable molecule:

1. In the DNA double helix, the nitrogenous bases are held together by extensive / numerous hydrogen bonds ;
2. Complementary base-pairing via two hydrogen bonds between adenine and thymine, and three hydrogen bonds between cytosine and guanine ;
3. Extensive hydrophobic interactions between the stacked bases, stabilise the structure of the double helix ;
4. The adjacent nucleotides within each polynucleotide strand are held together by strong covalent phosphodiester bonds, which are not easily broken. In this way, the integrity of the DNA base sequence is maintained ;
5. Proof-reading ability of DNA polymerase ensures DNA molecule is replicated correctly and accurately / nucleotide sequences are intact ;

(Note: Idea of “numerous/extensive” bonds must be mention at least once in this part for students to attain the full 5 marks for this section)

Why DNA may not be absolutely stable:

6. Affected by mutagens that damage DNA or introduce random mutations ;
7. For e.g. physical factors such as ionising radiation like gamma rays / X-rays which cause formation of chemically active ions in the cells which are capable of damaging and breaking DNA OR Absorption of UV light causes DNA to increase in energy level, causing damage to DNA double helix by creating kinks ;
8. For e.g. chemical factors (carcinogens) such as ethidium bromide that cause chemical changes in bases resulting in incorrect base pairing, results in insertion or deletion of base pair ;
9. Allow small changes in nucleotide sequences through point mutations, generating new alleles and hence genetic variation in a population ;

Survival of species:

10. With genetic variation, there is heritable variation for natural selection to act upon in a population ;
11. Due to different proteins/gene products generated that could confer selective advantage for some individuals ;
12. Ref to selection pressure, only individuals best able to adapt will survive and produce fertile and viable offspring ;
13. Passing down beneficial alleles from parents to offspring ;
14. If DNA totally stable, no new alleles / mutations / traits, which will reduce survival ability of the species ;
15. Ref to how DNA not being totally stable allows heterozygote advantage, e.g. heterozygotes / carriers of sickle-cell anaemia individuals are malaria-resistant (greater fitness) than homozygous dominant or homozygous recessive individuals ;

Conclusion:

16. Ref to continuation of a species and its continued evolution relies on a balance between accurate transmission of nucleotide sequences and the need for random change to provide the variation needed to allow continued evolution, depending on selection pressure ;

QUESTION 5

(a) Outline how variation can arise in any population of living organisms. [13]

In sexually reproducing population

- 1 Meiosis and fertilisation generates the genetic variation within a sexually reproducing population;
- 2 During prophase I, ***crossing over*** occurs between the non-sister chromatids of homologous chromosomes;
- 3 leading to new allelic combinations on a chromosome;
- 4 During metaphase I, ***independent assortment**** of ***homologous chromosomes**** occurs where the orientation of bivalents is random, as chromosomes line up along the metaphase plate;
- 5 leading to different chromosomal combinations in different gametes;
- 6 2^n different combinations of (chromosomes in) gametes, where n represents the haploid number of chromosomes in the species, can be obtained as a result of meiosis;
- 7 Random fusion / fertilisation of these gametes carrying different combinations of chromosomes adds to genetic variation of the zygote formed;

In asexual reproducing population (Bacteria)

Transformation

- 8 Bacterial cells take up foreign DNA from the surrounding medium via a cell surface receptor
- 9 Incorporating / integration of foreign DNA into its own DNA (via homologous recombination)

Transduction - Award marks either for generalized or specialised transduction only

- 10 A phage infects and injects its DNA into a bacterium and replicate in the cell.
- 11 Due to mistakes in the phage's reproductive cycle, a small piece of bacteria DNA was incorporated into the phage capsid
- 12 The resulting transducing phages infect other bacteria and newly infected cell acquires the original bacterial DNA / original bacterial DNA integrates into the DNA of the recipient

Conjugation

- 13 Attachment of F^+ and F^- bacterium via sex pilus made by F^+ cell. Sex pilus retracts, the two bacteria cells come into physical contact via conjugation tube/mating bridge
- 14 Single strand of F plasmid breaks at origin of transfer and move into recipient bacterium through conjugation tube

In both populations

- 15 Mutations occurs to generate new alleles reference to gene mutation or chromosomal mutation

(b) Discuss the suggestion that all living organisms on Earth depend on phosphate.
[12]

Cell membranes (max 4)

- 1 All living organisms have cell membranes require phosphate for phospholipids;
- 2 Phospholipids help to regulate what enter or leaves the cell or the organelles (idea of partial permeability);
- 3 Phospholipids contribute to the fluidity of cell membrane;
- 4 Allowing processes such as endocytosis / exocytosis to occur;
- 5 ref to phosphate being negatively charged allowing interaction with aqueous medium/ allowing bilayer to form;
- 6 allowing embedding of proteins in membrane;

Nucleic acids (max 3)

- 7 The genetic code depends on nucleic acids;
- 8 The backbone of nucleic acids is a sugar-phosphate chain / nucleic acid are made up of nucleotides comprise phosphate group, pentose sugar and nitrogenous base
- 9 Therefore without nucleic acids no storage of genetic information;

ATP (max 3)

- 10 Most energy are stored in the bonds of ATP; idea of ATP is energy currency within the cell;
- 11 The hydrolysis of the third phosphate group releases large amount of energy/ ATP more energy reach than ADP;
- 12 ref to use of ATP in a specific eg. (transcription, amino acid activation, active transport, endo- exocytosis, AVP)

Photosynthesis (max 3)

- 13 NADP is a coenzyme which can be reduced as it can carry high energy electrons and protons;
- 14 Oxidised NADP is the final proton and electron acceptor during non-cyclic photophosphorylation in light-dependent reaction;
- 15 Reduced NADP provides the reducing power to convert glycerate-3-phosphate to glyceraldehyde phosphate in the Calvin cycle;

Cell signaling (max 3)

- 16 When RTK dimerises, the tyrosine kinase function is activated resulting in autophosphorylation of tyrosine residues in the tails of the RTK, activating the receptor/ Activation of G-protein by GTP;
- 17 Each activated protein kinase will initiate a sequential phosphorylation and activation of other kinases, resulting in a phosphorylation cascade.
- 18 At each catalytic step in the cascade, the number of activated products is much greater than those in the preceding step resulting in signal amplification.
- 19 reference to cAMP as second messenger in signal transduction pathway;

AVP;

- 20 GTP used in translation;
- 21 GTP used in substrate level phosphorylation in Krebs cycle